

Similarity - Finding Sides Using Scale Factor

© FREEFALL MATHEMATICS - FREEFALL MATHEMATICS ALTITUDE BOOK 2 - LICENSED FOR NON-COMMERCIAL USE

Lets start with a quick review of equivalent ratios. Find the value of the pronumerals in the ratios.

1 $5 : 1 = 30 : x$
 $x = \underline{\hspace{2cm}}$

2 $4 : 3 = 16 : y$
 $y = \underline{\hspace{2cm}}$

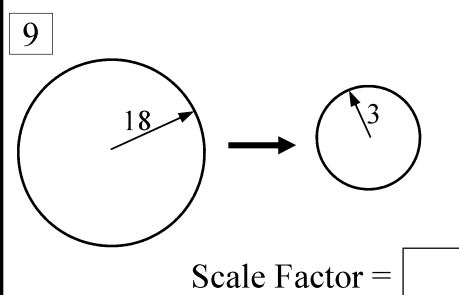
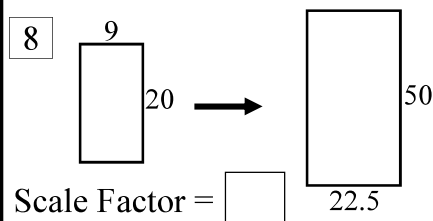
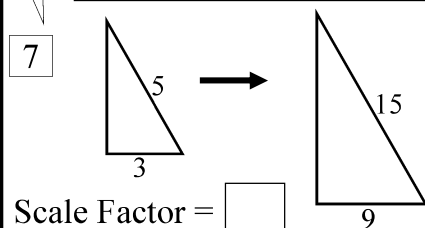
3 $3 : 2 = k : 22$
 $k = \underline{\hspace{2cm}}$

4 $a : 24 = 10 : 3$
 $a = \underline{\hspace{2cm}}$

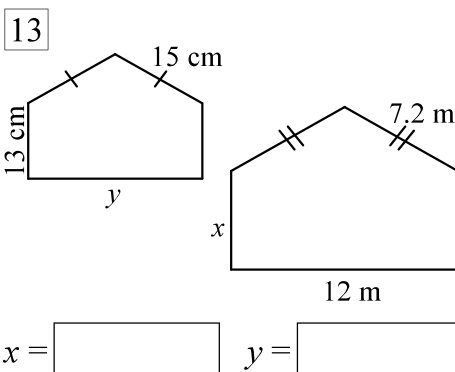
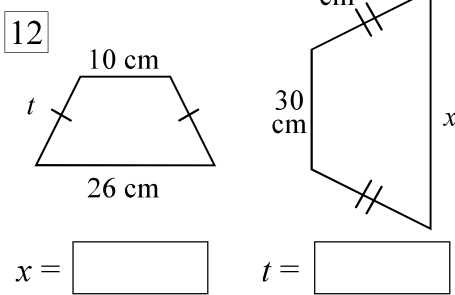
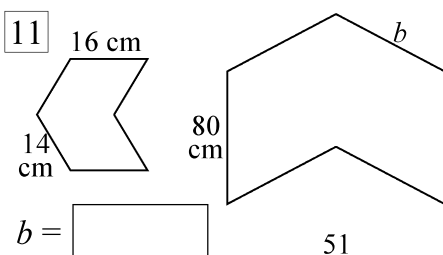
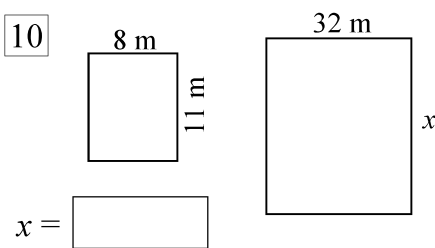
5 $20 : 7 : 13 = x : 35 : y$
 $x = \underline{\hspace{2cm}}$ $y = \underline{\hspace{2cm}}$

6 $b : 4 : 36 = 4 : 1 : f$
 $b = \underline{\hspace{2cm}}$ $f = \underline{\hspace{2cm}}$

Find the scale factor applied to the first shape to get the second.

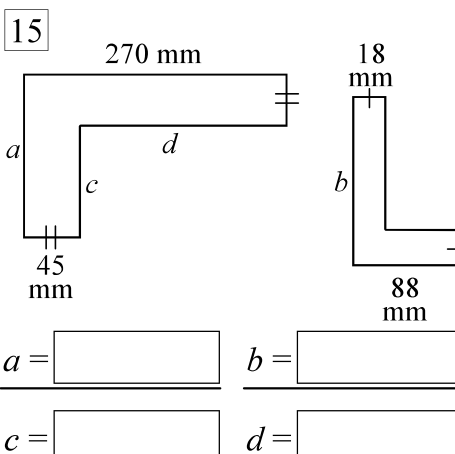


Use your ratio skills to find the unknown sides on these similar figures. Drawings not to scale.

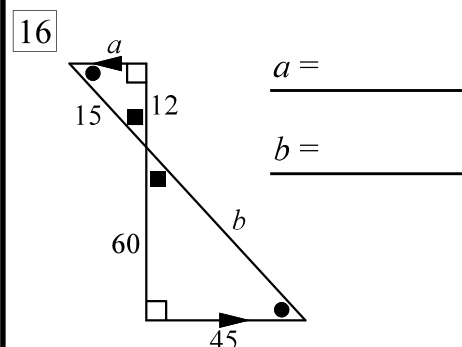


Find the perimeter of the larger shape

14

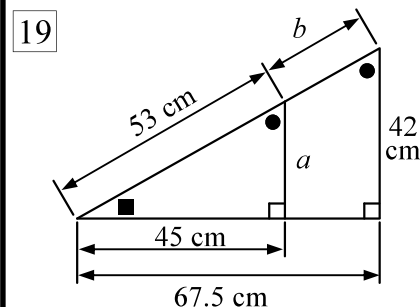


These triangles are similar. Find the value of a and b and give some angle theory reasoning.

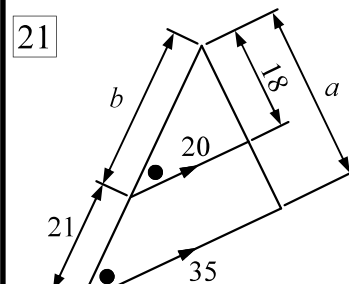


17 What angle property exists between the angles marked with a ● ?

18 What angle property exists between the angles marked with a ■ ?



20 Why are the angles marked with a ● equal?



22 Why are the angles marked with a ● equal?